

CUH601CMD Software Engineering Coursework Accompaniment

HOW TO MANAGE YOUR TEAM AND PROJECT**Intended Learning Outcome**

The objectives of CUH601CMD Software Engineering coursework are derived from the module objectives. Following the completion of the project, students are expected to be able to:

1. Evaluate an architectural design against accepted quality criteria.
2. Select, evaluate and use tools and techniques to successfully manage a large-scale software project, including configuration management and version control.
3. Examine software process models most suitable to develop a solution to a real-world problem.
4. Select, evaluate and apply standards, tools and techniques for assuring software quality.
5. Identify and engage with relevant social, professional, legal and ethical issues on a practical computing project.

The Project

The assessment for CUH601CMD Software Engineering is team-based. However, each team member will be assigned a relevant portion of individual tasks that will contribute to the completion of the team's project. Each team consists of 4 to 6 students, who will be working on the same project. The team must select one of the two study cases below.

Komodo Hub – A Digital Platform for Community-Supported Animal Conservation [Refer module AULA page]

JustInTime – FloatFry's Manufacturing Resource Planning [Refer module AULA page]

This module emphasizes a comprehensive software engineering approach to development. Therefore, the construction/coding aspect of the development of the proposed solution needs to be supported by equally strong evidence of practical project management and process control. To meet the objectives, students are asked to contribute to the creation of their team's comprehensive project report (~5000 words, citation excluded) covering three main aspects: **Project, Process, and Product**. Details on the required contributions from each role, the structure of the report document and what should be included in its content are available in the **HOW TO** documents provided on the module AULA page

Your Team

Your team is expected to complete a full cycle (scheduled for 8 weeks) of software development by strictly adopting Scrum. Initially, each member of your team will take on one of the following roles.

Product Owner
 Scrum Master
 (Developer-1) System Architect
 (Developer-2) Database Engineer
 (Developer-3) Web Developer
 (Developer-4) Cloud Engineer

Every week, your team will shuffle its members so that every student has the opportunity to take on the roles of Product Owner, Scrum Master, and Developer (either) at least once during the development. Your team must create good cohesion among its members so that changing roles will not have a negative impact on the project and the product. For role shuffling details and dates, please refer the assessment brief.

Please be advised that, although you will see teams working on the same study case may produce certain resemblances in work, their project report should be considerably different from each other as each team will go through different dynamics and propose different solutions.

Project Management

Your team will have sufficient number of weeks to complete the study case of your team's choice. During this time, your team must conduct 2 x 4-weeks Scrum Sprints that will be run in a sequence. The second Sprint should start immediately after the first one is completed. Please be advised that your team is only expected to implement weekly Scrums within each Sprint.

You will swap roles with another member of your team every week, and any role changes must be agreed upon by your team as a whole. Your new role will start effectively with the corresponding Scrum meeting. Please be advised that - applicable ONLY to the roles of Developer – your team does not have to follow the assignment of roles as mentioned above, i.e., your team can have four Web Developers or four Cloud Engineers in any week depending on the needs. An example of how to shuffle the roles can be seen below.

SPRINT 1	Week 1		SPRINT 1	Week 2		SPRINT 1	Week 3		SPRINT 1	Week 4
	Role			Role			Role			Role

Student 1	Product Owner		Student 1	Scrum Master		Student 1	Developer -3		Student 1	Developer - 3
Student 2	Scrum Master		Student 2	Product Owner		Student 2	Developer -4		Student 2	Developer - 4
Student 3	Developer-1		Student 3	Developer-1		Student 3	Product Owner		Student 3	Scrum Master
Student 4	Developer-2		Student 4	Developer-2		Student 4	Scrum Master		Student 4	Product Owner
Student 5	Developer-3		Student 5	Developer-3		Student 5	Developer-1		Student 5	Developer - 1
Student 6	Developer-4		Student 6	Developer-4		Student 6	Developer-2		Student 6	Developer - 2
SPRINT 2	Week 1		SPRINT 2	Week 2		SPRINT 2	Week 3		SPRINT 2	Week 4
	Role			Role			Role			Role
Student 1	Developer - 1		Student 1	Developer - 1		Student 1	Product Owner		Student 1	Scrum Master
Student 2	Developer - 2		Student 2	Developer - 2		Student 2	Scrum Master		Student 2	Product Owner
Student 3	Developer - 3		Student 3	Developer - 3		Student 3	Developer-1		Student 3	Developer-1
Student 4	Developer - 4		Student 4	Developer - 4		Student 4	Developer-2		Student 4	Developer-2
Student 5	Product Owner		Student 5	Scrum Master		Student 5	Developer-3		Student 5	Developer-3
Student 6	Scrum Master		Student 6	Product Owner		Student 6	Developer-4		Student 6	Developer-4

Furthermore, you can find below an example of project management and tasks scheduling. Please be advised that the tasks listed below are of general ones and functions as a guideline only. Your team may define new tasks and/or identify detailed activities to carry out whilst completing the tasks, which will consequently alter the schedule. You can also go into further detail by including activities listed in the [HOW TO Perform Your Role](#) document and assign a date and duration to each activity.

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